I hope your interview went well yesterday! Even if it felt like a marathon, practicing these types of coding challenges is the best way to stay sharp for the next one.

Here are **30 SQL coding questions** categorized by difficulty. I've focused on the logical "problems" you'd encounter in a technical screening.

Level 1: The Essentials (Easy)

Focus: `SELECT`, `WHERE`, `ORDER BY`, `LIMIT`

1. **Retrieve Unique Values:** Select all unique cities from a `customers` table.
2. **Filtering Patterns:** Find all employees whose names start with 'A' and end with 'n'.
3. **Basic Sorting:** List products with a price between $10 and $50, ordered from cheapest to most expensive.
4. **Handling Nulls:** Show all orders that do not have a `shipped_date` yet.
5. **Calculated Columns:** Return a list of employees with their full names (First + Last) and their annual salary (Monthly * 12).
6. **Top N Results:** Find the top 5 most expensive items in the `inventory` table.
7. **In-List Filtering:** Select all students who are enrolled in either 'Math', 'Science', or 'History'.
8. **Boolean Logic:** Find orders that were placed in 2023 but have a total amount less than $100.
9. **Date Extraction:** List all users who signed up in the month of February.
10. **Case Sensitivity:** Use `LOWER()` to find a specific email address regardless of how it was typed.

Level 2: Aggregations & Joins (Intermediate)

Focus: `GROUP BY`, `HAVING`, `JOIN`, `INNER/LEFT/RIGHT`

11. **Inner Join:** Show each order ID alongside the customer's name.
12. **Left Join:** List all departments and the number of employees in each (including departments with zero employees).
13. **Count & Group:** Count how many customers live in each country.
14. **Filtering Groups:** Find countries that have more than 10 customers using `HAVING`.
15. **Multiple Joins:** Connect `Orders`, `Order_Details`, and `Products` to show what items were bought in Order #101.
16. **Self-Join:** Given an `employees` table with a `manager_id`, list each employee and their manager's name.
17. **Average Calculation:** Find the average order value for each month in 2023.
18. **Max/Min:** Find the highest and lowest salary in each department.
19. **Sum with Conditions:** Calculate the total revenue generated only from 'Electronics' category products.
20. **Join with Filtering:** List customers who have never placed an order (Left Join where Foreign Key is NULL).

Level 3: Advanced Logic & Analytics (Hard)

Focus: `Subqueries`, `CTEs`, `Window Functions`, `CASE`

21. **Subquery in WHERE:** Find employees whose salary is higher than the company average.
22. **Correlated Subquery:** List products that are priced higher than the average price within their own category.
23. **Common Table Expressions (CTE):** Create a CTE to find the top 3 spenders, then join it with the `loyalty_program` table.
24. **RANK/DENSE_RANK:** Rank employees by salary within each department.
25. **Running Total:** Calculate a cumulative sum of sales for each day.
26. **Row Numbering:** Identify and select only the 2nd highest salary in the table.
27. **Conditional Logic:** Use `CASE` to label orders as 'Small' (<$50), 'Medium' ($50-$200), or 'Large' (>$200).
28. **Date Arithmetic:** Find all customers who haven't made a purchase in the last 6 months.
29. **Lead/Lag:** Compare today's sales to yesterday's sales using the `LAG()` function.
30. **String Manipulation:** Extract the domain name (e.g., 'gmail.com') from a column of email addresses.

Tips for SQL Success

Think in Sets: SQL isn't procedural like Python; it's about defining the "set" of data you want.

Order of Execution: Remember that `FROM` happens first, then `WHERE`, then `GROUP BY`, then `HAVING`, and `SELECT` happens almost last!

Would you like me to provide the solution code for a specific number from this list?